

**Course ID and Title:** CSCI 642 Robot Learning

**Units:** 4

**Term—Day—Time:** Fall 2026 — Wed — 3:30PM-6:50PM

**Location:** DMC 100

**Instructor:** Erdem Biyik

**Office Hours:** Wed 2:00PM-3:00PM @ GCS SB3 (LL2 Floor)

**Contact Info:** [biyik@usc.edu](mailto:biyik@usc.edu)

**Catalogue Description**

Interdisciplinary course on robot learning covers theory and practice: reinforcement learning, imitation learning, computer vision, sim-to-real transfer, multi-agent systems, safety, applications, and hands-on projects.

**Course Description**

Robot learning is an interdisciplinary field at the intersection of robotics, machine learning, cognitive science, and control theory, aiming to create intelligent and adaptable robotic systems capable of learning from their environment and experience. With rapid advances in artificial intelligence and computing power, as well as the possibility of having larger datasets, robot learning has the potential to revolutionize a wide range of applications, from manufacturing and healthcare to transportation and personal assistance. However, developing learning algorithms for real-world robotic systems poses unique challenges due to the complexities of the physical world, safety concerns, and the need for efficient and robust learning methods.

This course provides a comprehensive introduction to the fundamentals of robot learning, covering topics such as reinforcement learning, computer vision, meta-learning, sim-to-real transfer, and multi-agent learning. Students will explore cutting-edge techniques in imitation learning, inverse reinforcement learning, representation learning, and safe and robust learning, while also discussing the real-world applications and challenges of robot learning. The course is designed to be accessible to PhD students in robotics, control theory, machine learning, artificial intelligence, optimization, and related fields; with an emphasis on both theoretical foundations and practical applications.

In addition to lectures, the course features a series of student-led presentations on recent research papers and a course project, allowing students to gain hands-on experience with the latest advances in robot learning and explore emerging research topics. Through a combination of lectures, homework assignments, presentations, and project work, students will develop a deep understanding of robot learning techniques and their potential to transform the way we interact with and utilize robots in our everyday lives.

**Learning Objectives**

At the end of this course, students will be able to:

1. Apply fundamental and state-of-the-art techniques in robot learning to solve complex problems.
2. Analyze challenges in real-world robot learning applications and evaluate potential solutions.
3. Design learning algorithms for robotic systems with a focus on safety, robustness, and data efficiency.

4. Critique recent research papers in robot learning and demonstrate practical applications through hands-on projects.
5. Propose innovative research directions in robot learning that address real-world challenges and advance the field of robotics.
6. Collaborate across disciplines such as robotics, control theory, and machine learning to develop innovative solutions to complex robot learning problems.

**Prerequisite(s):** N/A

**Co-Requisite(s):** N/A

**Concurrent Enrollment:** N/A

**Recommended Preparation:** CSCI 445L: Introduction to Robotics or CSCI 545: Robotics, and fundamental concepts in machine learning on the level of CSCI 467: Introduction to Machine Learning

## Course Notes

Syllabus, homework, and other class information will be posted on the course website.

## Technological Proficiency and Hardware/Software Required

Students will be assumed to be proficient in at least one programming language (preferably Python).

## Required Readings and Supplementary Materials

**Required textbook:** There will be no required textbook.

**Readings:** The following conference and journal papers are the required readings. All of them are publicly available on the Internet.

- **Week 1:**

- Zhou, T., Brown, M., Snavely, N. and Lowe, D.G., 2017. Unsupervised learning of depth and ego-motion from video. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1851-1858).
- Dosovitskiy, A., Beyer, L., Kolesnikov, A. and Weissenborn, D., et al., 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International conference on learning representations*.

- **Week 2:**

- Chen, T., Kornblith, S., Norouzi, M. and Hinton, G., 2020, November. A simple framework for contrastive learning of visual representations. In *International conference on machine learning* (pp. 1597-1607). PMLR.
- Bobu, A., Liu, Y., Shah, R., Brown, D.S. and Dragan, A.D., 2023, March. Sirl: Similarity-based implicit representation learning. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 565-574).

- **Week 3:**

- Hafner, D., Lillicrap, T., Ba, J. and Norouzi, M., 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*.
- Haarnoja, T., Zhou, A., Abbeel, P. and Levine, S., 2018, July. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning* (pp. 1861-1870). PMLR.

- **Week 4:**

- Chen, L., Lu, K., Rajeswaran, A., Lee, K., et al., 2021. Decision transformer: Reinforcement learning via sequence modeling. *Advances in neural information processing systems*, 34, pp.15084-15097.
- Chi, C., Xu, Z., Feng, S., Cousineau, E., et al., 2025. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11), pp.1684-1704.

- **Week 5:**

- Ross, S., Gordon, G. and Bagnell, D., 2011, June. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics* (pp. 627-635). JMLR Workshop and Conference Proceedings.
- Ho, J. and Ermon, S., 2016. Generative adversarial imitation learning. *Advances in neural information processing systems*, 29.

- **Week 6:**
  - Florence, P., Lynch, C., Zeng, A., Ramirez, et al., 2022, January. Implicit behavioral cloning. In *Conference on robot learning* (pp. 158-168). PMLR.
  - Shafiullah, N.M., Cui, Z., Altanzaya, A.A. and Pinto, L., 2022. Behavior transformers: Cloning k modes with one stone. *Advances in neural information processing systems*, 35, pp.22955-22968.
- **Week 7:**
  - Hadfield-Menell, D., Milli, S., Abbeel, P., Russell, S.J., et al., 2017. Inverse reward design. *Advances in neural information processing systems*, 30.
  - Rafailov, R., Sharma, A., Mitchell, E., Manning, C.D., et al., 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36, pp.53728-53741.
- **Week 8:**
  - Kwon, M., Biyik, E., Talati, A., Bhasin, K., et al., 2020, March. When humans aren't optimal: Robots that collaborate with risk-aware humans. In *Proceedings of the 2020 ACM/IEEE international conference on human-robot interaction* (pp. 43-52).
  - Awad, E., Dsouza, S., Kim, R., Schulz, J., et al., 2018. The moral machine experiment. *Nature*, 563(7729), pp.59-64.
- **Week 9:**
  - Bajcsy, A., Losey, D.P., O'malley, M.K. and Dragan, A.D., 2017, October. Learning robot objectives from physical human interaction. In *Conference on robot learning* (pp. 217-226). PMLR.
  - Jeon, H.J., Losey, D.P., Sadigh, D., 2020, July. Shared Autonomy with Learned Latent Actions. In *Robotics: Science and Systems*.
- **Week 10:**
  - Peng, X.B., Andrychowicz, M., Zaremba, W. and Abbeel, P., 2018, May. Sim-to-real transfer of robotic control with dynamics randomization. In *2018 IEEE international conference on robotics and automation (ICRA)* (pp. 3803-3810). IEEE.
  - James, S., Wohlhart, P., Kalakrishnan, M., Kalashnikov, D., et al., 2019. Sim-to-real via sim-to-sim: Data-efficient robotic grasping via randomized-to-canonical adaptation networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12627-12637).
- **Week 11:**
  - Zintgraf, L., Shiarlis, K., Igl, M., Schulze, S., et al., 2020. VariBAD: A Very Good Method for Bayes-Adaptive Deep RL via Meta-Learning. In *International Conference on Learning Representations*.
  - Shridhar, M., Manuelli, L. and Fox, D., 2023, March. Perceiver-actor: A multi-task transformer for robotic manipulation. In *Conference on Robot Learning* (pp. 785-799). PMLR.
- **Week 12:**
  - Achiam, J., Held, D., Tamar, A. and Abbeel, P., 2017, July. Constrained policy optimization. In *International conference on machine learning* (pp. 22-31). PMLR.
  - Hsu, K.C., Hu, H. and Fisac, J.F., 2023. The safety filter: A unified view of safety-critical control in autonomous systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 7.
- **Week 13:**
  - Silver, D., Huang, A., Maddison, C.J., Guez, A., Sifre, L., Van Den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M. and Dieleman, S., 2016. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587), pp.484-489.
  - Foerster, J., Assael, I.A., De Freitas, N. and Whiteson, S., 2016. Learning to communicate with deep multi-agent reinforcement learning. *Advances in neural information processing systems*, 29.
- **Week 14:**
  - Brohan, A., Chebotar, Y., Finn, C., Hausman, K., et al., 2023, March. Do as i can, not as i say: Grounding language in robotic affordances. In *Conference on robot learning* (pp. 287-318). PMLR.
  - Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., et al., 2023, December. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning* (pp. 2165-2183). PMLR.

## Optional Readings and Supplementary Materials

The following books are recommended:

- [Reinforcement Learning: An Introduction](#) by Sutton and Barto

- Modern Adaptive Control and Reinforcement Learning (MACRL) by Bagnell, Boots, and Choudhury
- Principles of Robot Autonomy by Lorenzetti and Pavone

## Description of Assignments and How They Will Be Assessed

- **Readings:** Students will write summaries of the required readings every week. For each reading, the summary is expected to include 1-2 sentence description of the problem the paper is trying to solve, 2-3 sentences of the core novel ideas or technical contributions of the paper, 3-4 sentences of what specific techniques it uses, and 1-2 sentences of its main findings.
- **Homework:** Students will be assigned three homework sets that consist of both report questions and programming questions (in Python). Each student will select two of the three homework sets to complete. Report questions will require students to work on problems related to past lectures with pen and paper. Programming questions will require students to implement some of the methods covered in the lectures, occasionally with further improvements, and experiment them on simulated robot environments and/or machine learning tasks.
- **Class Presentation:** Students will present 1 research paper from the presentation papers list below. In the beginning of the semester, a sign-up sheet will be released for students to claim papers for presentations. After all students sign up, the presentations will be scheduled by the instructor to ensure presentations take place in the classes that are on the relevant topic. Presentations will be followed by open discussions. Students will be graded based on their presentations.
- **Midterm Exam:** One midterm exam will be taken during the scheduled class time. The students will individually answer the exam questions by pen/pencil on paper, without access to books or electronic devices. The teaching staff will make every effort to return graded exams within one week. Students will then have one week from the return date to raise any grading-related issues.
- **Course Project:** In addition, students will be required to work on a course project in groups of 2-3. The projects must have both robotics and machine learning components. They can be, for example, application-dependent improvements over an existing robot learning method, a novel robot learning related application of an existing technique, or a completely new method that may have potential benefits. Students will write a project proposal (2 pages excluding references), write a milestone report (2 pages excluding references), present their findings in an oral presentation (15 minutes excluding Q&A), and write a conference paper-style project report (6-8 pages excluding references). Instructor and teaching assistant(s) will provide feedback to the students on their proposal and milestone reports to guide progress.
- **Peer Review:** Students will be randomly assigned a final project report of another group. Each student will individually write a one-page peer review for that report. The students will be assessed based on the quality of the peer review they write. The reviews will be shared with the group members, but they will not affect the grade of the group.

**Presentation Papers List:** The following conference and journal papers are the papers students can choose to present. All of them are publicly available on the Internet.

- Oord, A.V.D., Li, Y. and Vinyals, O., 2018. Representation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748.
- Banayeeanzade, A., Bahrani, F., Zhou, Y. and Biyik, E., 2025. GABRIL: Gaze-Based Regularization for Mitigating Causal Confusion in Imitation Learning. In International Conference on Intelligent Robots and Systems (IROS).
- Roh, J., Desingh, K., Farhadi, A. and Fox, D., 2022, January. LanguageRefer: Spatial-language model for 3d visual grounding. In Conference on Robot Learning (pp. 1046-1056). PMLR.
- Pal, A., Qiu, Y. and Christensen, H., 2021, October. Learning hierarchical relationships for object-goal navigation. In Conference on Robot Learning (pp. 517-528). PMLR.
- Janner, M., Fu, J., Zhang, M. and Levine, S., 2019. When to trust your model: Model-based policy optimization. Advances in neural information processing systems, 32.
- Laskin, M., Srinivas, A. and Abbeel, P., 2020, November. Curl: Contrastive unsupervised representations for reinforcement learning. In International conference on machine learning (pp. 5639-5650). PMLR.
- Sontakke, S., Zhang, J., Arnold, S., Pertsch, K., Biyik, E., Sadigh, D., Finn, C. and Itti, L., 2023. Roboclip: One demonstration is enough to learn robot policies. Advances in Neural Information Processing Systems, 36, pp.55681-55693.

- Bahety, A., Mandikal, P., Abbatematteo, B. and Martín-Martín, R., 2024. Screwmimic: Bimanual imitation from human videos with screw space projection. In *Robotics: Science and Systems* (2024).
- Lepert, M., Fang, J. and Bohg, J., 2025. Phantom: Training robots without robots using only human videos. In *Conference on Robot Learning (CoRL)*.
- Kareer, S., Patel, D., Punamiya, R., Mathur, P., Cheng, S., Wang, C., Hoffman, J. and Xu, D., 2025, May. Egomimic: Scaling imitation learning via egocentric video. In *2025 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 13226-13233). IEEE.
- Ajay, A., Du, Y., Gupta, A., Tenenbaum, J., Jaakkola, T. and Agrawal, P., 2022. Is conditional generative modeling all you need for decision-making?. In *International Conference on Learning Representations*.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O., 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- Reddy, S., Dragan, A., Levine, S., 2018. Shared Autonomy via Deep Reinforcement Learning. In *Robotics: Science and Systems (RSS)*.
- Dass, S., Pertsch, K., Zhang, H., Lee, Y., Lim, J.J. and Nikolaidis, S., 2023, January. PATO: Policy Assisted TeleOperation for Scalable Robot Data Collection. In *Robotics: Science and Systems*.
- Zhao, M., Simmons, R. and Admoni, H., 2022, October. Coordination with humans via strategy matching. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 9116-9123). IEEE.
- Park, J., Seo, Y., Liu, C., Zhao, L., Qin, T., Shin, J. and Liu, T.Y., 2021. Object-aware regularization for addressing causal confusion in imitation learning. *Advances in Neural Information Processing Systems*, 34, pp.3029-3042.
- Li, X., Liu, M., Zhang, H., Yu, C., Xu, J., Wu, H., Cheang, C., Jing, Y., Zhang, W., Liu, H. and Li, H., 2024. Vision-Language Foundation Models as Effective Robot Imitators. In *The Twelfth International Conference on Learning Representations*.
- Goyal, A., Blukis, V., Xu, J., Guo, Y., et al., 2024. RVT-2: Learning Precise Manipulation from Few Demonstrations. In *Robotics: Science and Systems (RSS)*.
- Zhang, J., Luo, Y., Anwar, A., Sontakke, S.A., et al., 2025. ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations. In *Conference on Robot Learning (CoRL)*.
- Kumar, S., Zamora, J., Hansen, N., Jangir, R. and Wang, X., 2023, March. Graph inverse reinforcement learning from diverse videos. In *Conference on Robot Learning* (pp. 55-66). PMLR.
- Fu, M., Huang, H., Datta, G., Chen, L.Y., et al., 2025. ICRT: In-Context Imitation Learning via Next-Token Prediction. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*.
- Chan, A.J. and van der Schaar, M., 2021. Scalable Bayesian Inverse Reinforcement Learning. In *International Conference on Learning Representations*.
- Kamalaruban, P., Devidze, R., Cevher, V. and Singla, A., 2019. Interactive Teaching Algorithms for Inverse Reinforcement Learning. In *Twenty-Eighth International Joint Conference on Artificial Intelligence* (pp. 2692-2700). IJCAI.
- Ho, M.K., Littman, M., MacGlashan, J., Cushman, F. and Austerweil, J.L., 2016. Showing versus doing: Teaching by demonstration. *Advances in neural information processing systems*, 29.
- Bobu, A., Wiggert, M., Tomlin, C. and Dragan, A.D., 2022. Inducing structure in reward learning by learning features. *The International Journal of Robotics Research*, 41(5), pp.497-518.
- Korkmaz, Y., Biyik, E., et al., 2025. MILE: Model-Based Intervention Learning. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*.
- Liu, H., Nasiriany, S., Zhang, L., Bao, Z. and Zhu, Y., 2025. Robot learning on the job: Human-in-the-loop autonomy and learning during deployment. *The International Journal of Robotics Research*, 44(10-11), pp.1727-1742.
- Matas, J., James, S. and Davison, A.J., 2018, October. Sim-to-real reinforcement learning for deformable object manipulation. In *Conference on Robot Learning* (pp. 734-743). PMLR.
- Du, Y., Watkins, O., Darrell, T., Abbeel, P. and Pathak, D., 2021, May. Auto-tuned sim-to-real transfer. In *2021 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 1290-1296). IEEE.
- Nasiriany, S., Maddukuri, A., Zhang, L., Parikh, A., Lo, A., Joshi, A., Mandlekar, A. and Zhu, Y., 2024. Robocasa: Large-scale simulation of everyday tasks for generalist robots. *arXiv preprint arXiv:2406.02523*.

- Yu, T., Xiao, T., Stone, A., Tompson, J., Brohan, A., Wang, S., Singh, J., Tan, C., Peralta, J., Ichter, B. and Hausman, K., 2023. Scaling robot learning with semantically imagined experience. arXiv preprint arXiv:2302.11550.
- Chan, L., Critch, A. and Dragan, A., 2021. Human irrationality: both bad and good for reward inference. arXiv preprint arXiv:2111.06956.
- Julian, R., Swanson, B., Sukhatme, G., Levine, S., Finn, C. and Hausman, K., 2021, October. Never Stop Learning: The Effectiveness of Fine-Tuning in Robotic Reinforcement Learning. In Conference on Robot Learning (pp. 2120-2136). PMLR.
- Yoon, J., Kim, T., Dia, O., Kim, S., Bengio, Y. and Ahn, S., 2018. Bayesian model-agnostic meta-learning. Advances in neural information processing systems, 31.
- Sodhani, S., Zhang, A. and Pineau, J., 2021, July. Multi-task reinforcement learning with context-based representations. In International conference on machine learning (pp. 9767-9779). PMLR.
- Hu, H., Lerer, A., Peysakhovich, A. and Foerster, J., 2020, November. "Other-Play" for Zero-Shot Coordination. In International Conference on Machine Learning (pp. 4399-4410). PMLR.
- Gervet, T., Xian, Z., Gkanatsios, N. and Fragkiadaki, K., 2023, December. Act3D: 3D Feature Field Transformers for Multi-Task Robotic Manipulation. In Conference on Robot Learning (pp. 3949-3965). PMLR.
- Ke, T.W., Gkanatsios, N. and Fragkiadaki, K., 2025, January. 3D Diffuser Actor: Policy Diffusion with 3D Scene Representations. In Conference on Robot Learning (pp. 1949-1974). PMLR.
- Sui, Y., Gotovos, A., Burdick, J. and Krause, A., 2015, June. Safe exploration for optimization with Gaussian processes. In International conference on machine learning (pp. 997-1005). PMLR.
- Robey, A., Hu, H., Lindemann, L., Zhang, H., Dimarogonas, D.V., Tu, S. and Matni, N., 2020, December. Learning control barrier functions from expert demonstrations. In 2020 59th IEEE Conference on Decision and Control (CDC) (pp. 3717-3724). IEEE.
- Bansal, S. and Tomlin, C.J., 2021, May. Deepreach: A deep learning approach to high-dimensional reachability. In 2021 IEEE International Conference on Robotics and Automation (ICRA) (pp. 1817-1824). IEEE.
- Huang, W., Wang, C., Li, Y., Zhang, R. and Fei-Fei, L., 2025, January. ReKep: Spatio-Temporal Reasoning of Relational Keypoint Constraints for Robotic Manipulation. In Conference on Robot Learning (pp. 4573-4602). PMLR.
- Singh, I., Blukis, V., Mousavian, A., Goyal, A., Xu, D., Tremblay, J., Fox, D., Thomason, J. and Garg, A., 2023, May. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. In 2023 IEEE International Conference on Robotics and Automation (ICRA) (pp. 11523-11530). IEEE.
- Shi, L.X., Hu, Z., Zhao, T.Z., Sharma, A., Pertsch, K., Luo, J., Levine, S. and Finn, C., 2024, January. Yell At Your Robot: Improving On-the-Fly from Language Corrections. In Robotics: Science and Systems.
- Kim, M.J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E.P., Sanketi, P.R., Vuong, Q. and Kollar, T., OpenVLA: An Open-Source Vision-Language-Action Model. In 8th Annual Conference on Robot Learning.
- Duan, J., Yuan, W., Pumacay, W., Wang, Y.R., Ehsani, K., Fox, D. and Krishna, R., 2025, January. Manipulate-Anything: Automating Real-World Robots using Vision-Language Models. In Conference on Robot Learning (pp. 5326-5350). PMLR.
- Snyder, D., Hancock, A.J., Badithela, A., Dixon, E., et al., 2025. Is your imitation learning policy better than mine? Policy comparison with near-optimal stopping. In Robotics: Science and Systems (RSS).
- Badithela, A., Snyder, D., Zha, L., Mikhail, J., O'Kelly, M., Dixit, A. and Majumdar, A., 2025. Reliable and Scalable Robot Policy Evaluation with Imperfect Simulators. arXiv preprint arXiv:2510.04354.
- Liu, F., Fang, K., Abbeel, P. and Levine, S., 2024, March. Moka: Open-vocabulary robotic manipulation through mark-based visual prompting. In First Workshop on Vision-Language Models for Navigation and Manipulation at ICRA 2024.
- Shridhar, M., Manuelli, L. and Fox, D., 2023, March. Perceiver-actor: A multi-task transformer for robotic manipulation. In Conference on Robot Learning (pp. 785-799). PMLR.
- Zhao, T.Z., Kumar, V., Levine, S. and Finn, C., 2023. Learning fine-grained bimanual manipulation with low-cost hardware. In Robotics: Science and Systems..

## Grading Breakdown

| Assessment Tool (assignments) | % of Grade |
|-------------------------------|------------|
| Readings                      | 7          |
| Homework                      | 20         |
| Class Presentation            | 15         |
| Midterm Exam                  | 15         |
| Course Project                | 38         |
| - Proposal (5%)               |            |
| - Milestone Report (8%)       |            |
| - Presentation (10%)          |            |
| - Final Project Report (15%)  |            |
| Peer Review                   | 5          |
| <b>TOTAL</b>                  | <b>100</b> |

## Grading Scale

Numeric scores will be converted to letter grades based on both individual performance trends and the overall class performance. Grade boundaries will be adjusted as needed to account for the difficulty of assessments and ensure fairness.

## Assignment Submission Policy

Homework reports and codes will be submitted online with the deadlines listed on the course schedule.

## Course-Specific Policies

N/A.

## Attendance

Attendance to the lectures is expected, but not graded.

## Academic Integrity for this Class

Unless otherwise noted, this course will follow the expectations for academic integrity as stated in the [USC Student Handbook](#). The general USC guidelines on Academic Integrity and Course Content Distribution are provided later in this syllabus.

For this class,

- Group work: All homework assignments are expected to be completed individually. Course project is done in groups unless instructor permission is given for individual projects.
- Computer programs: Plagiarism includes the submission of code written by, or otherwise obtained from someone else.

If found responsible for an academic violation, students may be assigned university outcomes, such as suspension or expulsion from the university, and grade penalties, such as an “F” grade on the assignment, exam, and/or in the course.

Please ask the instructor if you are unsure about what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Class Recordings and Course Content Distribution: You may not record this class without the express permission of the instructor and all other students in the class. Distribution of any notes, recordings, exams, or other materials from a university class or lectures — other than for individual or class group study — is prohibited without the express permission of the instructor; violations will be considered an intentional act to facilitate or enable academic dishonesty and reported to the university.

### Use of Generative AI in this Course

In this course, you are permitted to use artificial intelligence (AI)-powered programs to help you, but only on assignments that explicitly indicate a permitted use of AI. However:

- You should also be aware that AI text generation tools may present incorrect information, biased responses, and incomplete analyses; thus, their answers may not meet the standards of this course.
- To adhere to our university values, you must cite any AI-generated material (e.g., text, images, and other content) included or referenced in your work and provide the prompts used to generate the content. Using an AI tool to generate content without proper attribution will be treated as plagiarism and reported to the Office of Academic Integrity.

Please review the instructions in each assignment for more details on how and when to use AI Generators for your submissions.

### Course Evaluations

In addition to the official end-of-the semester course evaluations, students will be given a chance to send anonymous evaluations/feedback in the middle of the semester.

### Course Schedule

For each topic listed below, the class will involve a comprehensive lecture provided by the instructor, possibly followed by the students' class presentations depending on the presentation assignments.

|                | Topics                                                                  | Deliverables                                        |
|----------------|-------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Week 1</b>  | Basics of robotics<br>Fundamentals of machine learning                  |                                                     |
| <b>Week 2</b>  | Basics of computer vision for robotics<br>Representation learning       | Homework #1 is out                                  |
| <b>Week 3</b>  | Dynamic programming<br>Tabular reinforcement learning                   |                                                     |
| <b>Week 4</b>  | Model-based reinforcement learning<br>Model-free reinforcement learning | Deadline: Homework #1                               |
| <b>Week 5</b>  | Imitation learning                                                      | Homework #2 is out                                  |
| <b>Week 6</b>  | Inverse reinforcement learning                                          |                                                     |
| <b>Week 7</b>  | Learning from human feedback                                            | Deadline: Homework #2<br>Deadline: Project proposal |
| <b>Week 8</b>  | Human-in-the-loop robot learning                                        | Homework #3 is out                                  |
| <b>Week 9</b>  | Sim-to-real transfer<br>Midterm Exam                                    |                                                     |
| <b>Week 10</b> | Meta learning<br>Multi-task learning                                    | Deadline: Homework #3                               |
| <b>Week 11</b> | Multi-agent learning                                                    | Deadline: Project milestone report                  |
| <b>Week 12</b> | <i>No lecture</i>                                                       |                                                     |
| <b>Week 13</b> | Robot learning using natural language                                   |                                                     |
| <b>Week 14</b> | <i>No lecture</i>                                                       |                                                     |
| <b>Week 15</b> | Project Presentations<br>Conclusions                                    | Deadline: Final project report                      |
| <b>FINAL</b>   |                                                                         | Deadline: Peer review                               |



## **Academic Integrity**

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct — which includes any act of dishonesty in the production or submission of academic work (either in draft or final form) — is in contrast to the university’s mission to educate students through a broad array of academic, professional, and extracurricular programs.

This course will follow the expectations for academic integrity as stated in the [USC Student Handbook](#). All students are expected to submit assignments that are their own original work and prepared specifically for this course and section in this academic term. You may not submit work written by others or “recycle” work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of engaging in academic misconduct will be reported to the Office of Academic Integrity.

Other violations of academic misconduct include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

Academic dishonesty has a far-reaching impact and is considered a serious offense against the university. Violations will result in a grade penalty, such as a failing grade on the assignment or in the course, and disciplinary action from the university itself, such as suspension or even expulsion.

For more information about academic integrity see the [student handbook](#) or the [Office of Academic Integrity’s website](#), and university policies on [Research and Scholarship Misconduct](#).

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment or what information requires citation and/or attribution.

## **Course Content Distribution and Synchronous Session Recordings Policies**

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Student Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on the academic freedom of other students as well as the instructor. ([Living our Unifying Values: The USC Student Handbook](#), page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposes other than individual or group study. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which had been distributed to students or in any way had been displayed for use in relation to the class, whether obtained in class, via email, on the internet, or via any other media. Distributing course material without the instructor’s permission will be presumed to be an intentional act to facilitate or enable academic dishonesty and is strictly prohibited. ([Living our Unifying Values: The USC Student Handbook](#), page 13).

## Statement on University Academic and Support Systems

### Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. [The Office of Student Accessibility Services](#) (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at [osas.usc.edu](https://osas.usc.edu). You may contact OSAS at (213) 740-0776 or via email at [osasfrontdesk@usc.edu](mailto:osasfrontdesk@usc.edu).

### Student Financial Aid and Satisfactory Academic Progress:

To be eligible for certain kinds of financial aid, students are required to maintain Satisfactory Academic Progress (SAP) toward their degree objectives. Visit the [Financial Aid Office webpage](#) for [undergraduate](#)- and [graduate-level](#) SAP eligibility requirements and the appeals process.

### Support Systems:

[Counseling and Mental Health](#) - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

[988 Suicide and Crisis Lifeline](#) - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline consists of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

[Relationship and Sexual Violence Prevention Services \(RSVP\)](#) - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

[Office for Equity, Equal Opportunity, and Title IX \(EEO-TIX\)](#) - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

[Reporting Incidents of Bias or Harassment](#) - (213) 740-2500

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

[The Office of Student Accessibility Services \(OSAS\)](#) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

[USC Campus Support and Intervention](#) - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

[Diversity, Equity and Inclusion](#) - (213) 740-2101

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

[USC Emergency](#) - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

[USC Department of Public Safety](#) - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

[Office of the Ombuds](#) - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

[Occupational Therapy Faculty Practice](#) - (323) 442-2850 or [otfp@med.usc.edu](mailto:otfp@med.usc.edu)

Confidential Lifestyle Redesign services for USC students to support health-promoting habits and routines that enhance quality of life and academic performance.